

2520090058 PRACTICAL WEEK 5:

Implement a producer-consumer communication system using anonymous pipes where the parent process generates data and the child process consumes it. Measure the communication efficiency.

CODE:

```

1  UW PID0 5.89
2  File: kn.c
3
4  #include <stdio.h>
5  #include <stdlib.h>
6  #include <unistd.h>
7  #include <sys/wait.h>
8  #include <time.h>
9
10 #define N 100000
11
12 int main()
13 {
14     int fd[2];
15     pid_t pid;
16     int data;
17     long long sum = 0;
18
19     struct timespec start, end;
20
21     // Create anonymous pipe
22     if (pipe(fd) == -1)
23     {
24         perror("pipe failed");
25         exit(1);
26     }
27
28     // Start timer
29     clock_gettime(CLOCK_MONOTONIC, &start);
30
31     pid = fork();
32
33     if (pid < 0)
34     {
35         perror("fork failed");
36         exit(1);
37     }
38
39     if (pid > 0)
40     {
41         // =====
42         // PARENT - PRODUCER
43         // =====
44         close(fd[0]); // Close reading end
45
46         for (int i = 1; i <= N; i++)
47         {
48             data = i;
49
50             if (write(fd[1], &data, sizeof(data)) != sizeof(data))
51             {
52                 perror("write failed");
53                 exit(1);
54             }
55
56             close(fd[1]); // Signal end of data
57
58             // Wait for consumer
59             wait(NULL);
60
61             clock_gettime(CLOCK_MONOTONIC, &end);
62
63             double time_taken =
64                 (end.tv_sec - start.tv_sec) +
65                 (end.tv_nsec - start.tv_nsec) / 1e9;
66
67             double throughput =
68                 (N * sizeof(int)) / time_taken;
69
70             printf("\nProducer: Generated %d integers\n", N);
71         }
72     }
73 }

```

OUTPUT:

```

Last login: Sat Sep 11 09:42:08 on ttys000
screen@hillreddy@Green-MacBook-Pro ~ % nano kn.c
screen@hillreddy@Green-MacBook-Pro ~ % gcc kn.c -o kn
screen@hillreddy@Green-MacBook-Pro ~ % ./kn
Consumer: Received 100000 integers
Consumer: Sum = 5000050000
Producer: Generated 100000 integers
Communication time: 0.091774 seconds
Throughput: 4308532.72 bytes/second
Throughput: 4.16 MB/second
screen@hillreddy@Green-MacBook-Pro ~ % nano kn.c
screen@hillreddy@Green-MacBook-Pro ~ % █

```

Develop a program that executes the equivalent of the shell command: `ls -l | grep ".c"` using `fork()`, `pipe()`, `dup2()`, and `exec()` system calls.

OUTPUT:

```
264.c
3.c
31.e
322.c
334.c
34.c
35.c.save
399.c
4.c
4.c.save
44.c
45.c
455.c
457.c
467.c
46.c
79.c.save
a.c
ad.c
Applications
Applications (Parallels)
bb.c
bw.c
c.c
cc
cc.c
cv.c
d66.c
Documents
file.c
file.c.save
gm.c
gt.c
hl.c
la.c
ja.co
kk.c
kk.c.save
kk.c.save
kl.c.save
km.c
kn.c
l.c
la.c
ll.c.save
llll.c
m.c
mk.c.save
mn.c.save
Music
n.c.save
nmo.c.save
nlk.c.save
Pictures
process
process.c
process.c.save
projects
py.c
Public
pwl.c
pwl.c.save
q.c
source.txt
task.c
tt.c.save
v.c
wp.c
week2.c
week2kill.c
week3kill.c
ww.c
streamkhilreddy@Stream-MacBook-Pro ~ % nano cv.c
streamkhilreddy@Stream-MacBook-Pro ~ %
```